

## Lab 05 – Loops and Iterations

**CLO Alignment:** CLO2

### Learning Objectives

Students can use:

- for loop
  - while loop
  - break
  - continue
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### Exercise 1 – Multiplication Table

#### Problem Statement

Input:

5

Output:

5 x 1 = 5

5 x 2 = 10

...

5 x 10 = 50

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### Exercise 2 – Prime Number Detector

#### Problem Statement

Determine whether a number is prime.

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### Exercise 3 – ATM Withdrawal Simulator

#### Problem Statement

Initial balance:

1000

Loop until user exits.

Menu:

1. Withdraw
  2. Check balance
  3. Exit
- 

### Exercise 3 - Refactor Lab 3 - 5 using functions

Refactor Lab 3–5 using functions:

```
def loan_calculator():
```

```
def scholarship_checker():
```

```
def atm_system():
```

and integrate into a menu-driven application:

===== PYTHON MINI SYSTEM =====

1. Financial Tools
  2. Student Tools
  3. Authentication Tools
  4. Exit
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